Statistics
Syllabus & Homework

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CH05.01_28

28. Which Key Fits? A key ring contains four office keys that are identical in appearance, but only one will open your office door. Suppose you randomly select one key and try it. If it does not fit, you randomly select one of the three remaining keys. If that key does not fit, you randomly select one of the last two. Each different sequence that could occur in selecting the keys represents a set of equally likely simple events.

a. List the simple events in S and assign probabilities to the simple events.

b. Let x equal the number of keys that you try before you find the one that opens the door ($x = 1, 2, 3, 4$). Then assign the appropriate value of x to each simple event.

c. Calculate the values of $p(x)$ and display them in a table.

d. Construct a probability histogram for $p(x)$.

CH05.01_34

34. FDA Testing The maximum patent life for a new drug is 17 years. Subtracting the length of time required by the FDA for testing and approval of the drug provides the actual patent life of the drug—that is, the length of time that a company has to recover research and development costs and make a profit. Suppose the distribution of the lengths of patent life for new drugs is as shown here:

Years, x	3	4	5	6	7	8	9	10	11	12	13
$p(x)$	.03	.05	.07	.10	.14	.20	.18	.12	.07	.03	.01

a. Find the expected number of years of patent life for a new drug.

b. Find the standard deviation of x.

c. Find the probability that x falls into the interval $\mu \pm 2\sigma$.

CH05.02_44

44. McDonald's or Burger King? Suppose that 50% of all young adults prefer McDonald's to Burger King when asked to state a preference. A group of 10 young adults were randomly selected and their preferences recorded.

a. What is the probability that more than 6 preferred McDonald's?

b. What is the probability that between 4 and 6 (inclusive) preferred McDonald's?

c. What is the probability that between 4 and 6 (inclusive) preferred Burger King?

CH05.02_47

47. Plant Genetics A peony plant with red petals was crossed with a peony plant having streaky petals. The probability that an offspring from this cross has red flowers is .75. Let x be the number of plants with red petals resulting from 10 seeds from this cross that were collected and germinated.

a. Does the random variable x have a binomial distribution? If not, why not? If so, what are the values of n and p?

b. Find $P(x \geq 9)$.

c. Find $P(x \leq 1)$.

d. Would it be unusual to observe one plant with red petals and the remaining nine plants with streaky petals? If these experimental results actually occurred, what conclusions could you draw?

CH06.02_51

Human Heights *Assume that the heights of American men are normally distributed with a mean of 176.5 centimeters and a standard deviation of 8.9 centimeters. Use this information to answer the questions in Exercises 49–52.*

51. President Donald Trump is 1.88 m. Is this an unusual height?

Barron Trump › Height

206 cm

Image: Win McNamee, Getty Images

CH06.02_59

59. Gestation Times *The Biology Data Book* reports that the gestation time for human babies averages 278 days with a standard deviation of 12 days.[3] Suppose that these gestation times are normally distributed.

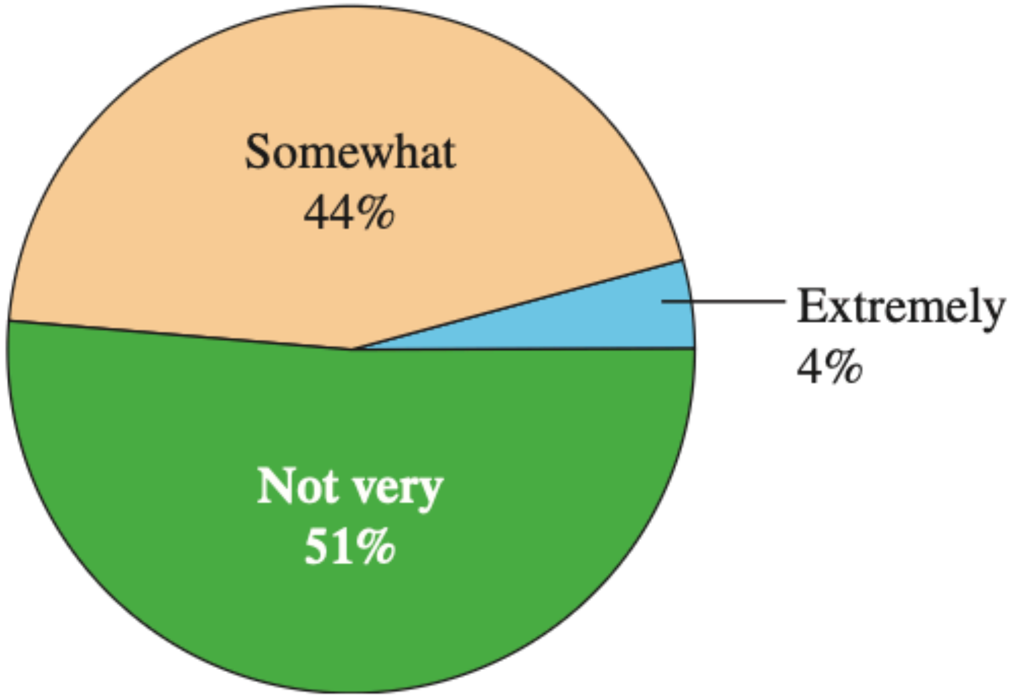
a. Find the upper and lower quartiles for the gestation times.

b. Would it be unusual to deliver a baby after only 6 months of gestation? Explain.

CH06.03_17

17. Cell Phone Etiquette A Snapshot in *USA Today* indicates that 51% of Americans say the average person is not very considerate of others when talking on a cell phone.[10] Suppose that 100 Americans are randomly selected. Find the approximate probability that 60 or more Americans would indicate that the average person is not very considerate of others when talking on a cell phone.

How Polite Are Cell Phone Users?



CH06.03_20

20. Genetic Defects Data indicate that a particular genetic defect occurs in 1 of every 1000 children. The records of a medical clinic show $x = 60$ children with the defect in a total of 50,000 examined.

a. If the 50,000 children were a random sample from the population of children represented by past records, what is the probability of observing a value of x equal to 60 or more?

b. Would you say that the observation of $x = 60$ children with genetic defects represents a rare event?

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